CSE310: Compiler Sessional, January 2026 Term
ICG Online Assignment Practice
Absolute Value Built-in abs()

Time: 40 Mins **Total Marks: 10**

Problem

Extend the expression grammar to support a built-in `abs()` function. It takes a single expression and returns its absolute (positive) value.

```
int a; a = -15; println(abs(a));
```

Task

- Modify the `factor` rule to support `abs(expression)`.
- Update the ICG visitor. After evaluating the inner expression, check if `EAX` is less than 0. If it is, negate `EAX` using the `NEG` instruction.

Mark Distribution

- Extending the grammar. (4)
- Assembly generation for conditional negation. (6)

Sample Input-Output

Input Program	Generated Sample Assembly (print library omitted)

<table>
<tr>
<td><pre>int main() {
    int a;
    a = 0 - 5;
    println(abs(a));
    println(abs(10));
    return 0;
}</pre></td>
<td><pre>main:
    PUSH EBP
    MOV EBP, ESP
    SUB ESP, 4
    MOV EAX, 0
    MOV EDX, EAX
    MOV EAX, 5
    SUB EDX, EAX
    MOV EAX, EDX
    MOV [EBP-4], EAX
    MOV EAX, [EBP-4]
    CMP EAX, 0
    JGE .L1
    NEG EAX
.L1:
    CALL print_number
    MOV EAX, 10
    CMP EAX, 0
    JGE .L2
    NEG EAX
.L2:
    CALL print_number
    MOV EAX, 0
    JMP .L3
.L3:
    ADD ESP, 4
    POP EBP
    RET</pre></td>
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<tr>
<td>Expected Output</td>
<td><pre>5
10</pre></td>
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